

# HAOPENG DENG

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## EDUCATION

**Guangzhou Maritime University**

**09/2022-07/2026**

B. Eng. in *Engineering Management* | School of Future Transportation

Thesis: Attribution Analysis of Digital Retrofitting Effectiveness in Urban Transportation Infrastructure

## ACADEMIC VISITS

**Massachusetts Institute of Technology**

**06/2026-09/2026**

Selected as one of 41 global participants in *MIT-UF-NEU Summer Research Camp* at Prof. Cathy Wu's Lab, collaborating with Ryan Hardesty Lewis on distilling interaction preferences for mixed-autonomy driving.

**Tsinghua University & Technical University of Munich**

**12/2025**

Selected as one of 67 global participants for the "*Smart Sustainable Cities and Transportation*" Winter School, completing 11 modules covering data-driven modeling, mixed-traffic systems, and urban computing.

## CONFERENCE PAPERS

- [C1] **Deng, H.**, He, S., Wang, D. (2026). FLAT: Smoothing the Rugged Landscape for Learnable, Sample-Efficient Traffic Calibration. Submitted to *The 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*.
- [C2] **Deng, H.**, Zheng, F., Xia, X.. (2026). SpinFlow: A Physics-Informed Spin Field Framework for Traffic Phase Inference and Transition Detection. Accepted in *The 29th IEEE International Conference on Intelligent Transportation Systems (ITSC)*. DOI: [10.48550/arXiv.2605.23306](https://doi.org/10.48550/arXiv.2605.23306).
- [C3] **Deng, H.**, Zheng, F., Ma, K., et al. (2025). Adaptive Truncated Schatten Norm for Traffic Data Imputation with Complex Missing Patterns. *The 25th COTA International Conference of Transportation Professionals (CICTP)*. American Society of Civil Engineers. (**Best Poster Award**). DOI: [10.1061/9780784486269.009](https://doi.org/10.1061/9780784486269.009).
- [C4] Xia, X., **Deng, H.\***, Ye, Y. (2026). Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation. Accepted in *The 26th CICTP Conference*.

## JOURNAL ARTICLES

- [J1] Xia, X., **Deng, H.**, Tang, J., et al. (2025). Dynamic Traffic Assignment under Mixed Traffic: Modeling, Evolution, and Solution Approaches. *Journal of Transport Science and Engineering*. (**Cover Paper, Oral Presented on AIAT 2026**). (Chinese). DOI: [10.16544/J.CNKI.CN43-1494/U.20251007001](https://doi.org/10.16544/J.CNKI.CN43-1494/U.20251007001).
- [J2] **Deng, H.**, Ma, K., Zhan, L., et al. (2025). Structural Equation Modeling of the Relationships among Traffic Safety KAP in Small-Medium Cities. *Journal of Transportation Engineering*. (Chinese). DOI: [10.13986/J.CNKI.JOTE.2026.04.002](https://doi.org/10.13986/J.CNKI.JOTE.2026.04.002).
- [J3] **Deng, H.**, Tang, J., Zhan, L., et al. (2025). Research on Traffic Governance Optimization in Kaiping City Based on a GRA-FCE Integrated Framework. *Journal of Transportation Engineering*. (Chinese). DOI: [10.13986/J.CNKI.JOTE.2026.04.011](https://doi.org/10.13986/J.CNKI.JOTE.2026.04.011).
- [J4] Wang, Y., Li, L., **Deng, H.**, et al. (2026). Sparse Stochastic Optimal Control under Control-Dependent Diffusion and Time-Varying Dynamics. Accepted in *Journal of Dynamical and Control Systems*.

## RESEARCH EXPERIENCES

**Guangdong Provincial Special Fund for Science and Technology Innovation Strategy**

**07/2024-04/2026**

Title: *Cooperative Game-Based Network Traffic Assignment Mechanisms for Enhancing Urban Transportation Efficiency* | Grant No. PDJH2024A289 | Project Leader | Advisor: Dr. Hu Lei

- Authored a systematic review on dynamic traffic assignment (DTA), synthesizing insights from 120+ studies on multi-class vehicle modeling, nonlinear equilibrium evolution, and solution algorithms<sup>[1]</sup>.
- Developed a VI-based DTA model that unifies UE-SO equilibria into a tunable spectrum, by decoupling CAV penetration into automation levels (AL) and connectivity levels (CL) as evolution variables.
- Proposed the AMPAS algorithm for solving DUE, DSO, and UE-SO mixed equilibria within the AL-CL framework, achieving 27.2% faster convergence than gradient projection on medium-scale networks.
- Directed team task delegation, milestone tracking, and mentoring of lower year members; coordinate regular progress meetings to align efforts and ensure on-schedule research deliverables.

**Guangdong Provincial College Student Innovation & Entrepreneur Training Program**

**11/2024-02/2026**

Title: *Intelligent Recognition of Urban Traffic Equilibrium States under Air-Ground Collaborative Monitoring* | Project Co-Leader | Instructor: Prof. Xinhai Xia

- Integrating heterogeneous data sources from roadside sensors, probe vehicles, and UAVs to construct high-dimensional traffic flow tensors, supporting accurate traffic equilibrium state recognition.
- Formulated three representative real-world data loss structures—element-wise, fiber-wise, and mixed—within urban traffic flow tensors, serving as a rigorous testbed for refining data recovery algorithms.
- Developed an adaptive truncated Schatten norm-based tensor completion model to effectively fill gaps in traffic datasets with complex missing patterns, improving MAPE by 10.6% and RMSE by 6.1%<sup>[C3]</sup>.
- Facilitated industry-academia partnership for traffic survey equipment and training, guided junior students in fieldwork to create datasets for analysing traffic flow dynamics and defining equilibrium states.

### **High-Quality Development Project for Hundred Counties, Thousand Towns, Myriad Villages 07/2024**

Title: *Investigation on Traffic Safety and Planning in Kaiping City* | Project Leader | Advisor: Dr. Hu Lei

- Collaborated with the Municipal Transportation Bureau to analyse traffic accident and violation data, identifying crash hotspots and associated hazard types, and designing two targeted questionnaires.
- Led a seven-person team to conduct on-site surveys, interviews, inspections, and drone photography, collecting data on traffic flow, vehicle management, pedestrian movement, and infrastructure.
- Utilised SPSS and LLM tools for analysis of survey data and interview transcripts, uncovering key issues and generating actionable insights to inform improvements in urban traffic safety and planning.
- Authored two papers—one on traffic safety behaviour, attitude and cognition relationship using Structural Equation Modeling<sup>[J2]</sup>, and another on the city's traffic governance assessment via Grey Relational and Fuzzy Evaluation<sup>[J3]</sup>; a synthesized report was later adopted by the Municipal Transportation Bureau.

## **INTERNSHIP**

### **Guangzhou Xueshujia Software Technology Co., Ltd.**

**01/2025-05/2025**

*Simulation Assistant Engineer* | Mentor: Dr. Wenchao Shen

- Deployed a multi-agent reinforcement learning module for CAV control in the company's traffic simulator, enhancing its capability to model and visualise V2X-based highway shock-wave propagation.
- Benchmarked the simulator against leading platforms such as SUMO using identical real-world trajectory datasets, diagnosing fidelity gaps and recommending optimisations to enhance competitiveness.
- Surveyed major traffic-related competitions and conferences over the past five years, analysing emerging trends in the field and discovering potential opportunities for industry-academia collaboration.

## **EXTRACURRICULAR ACTIVITIES**

### **Innovation and Entrepreneurship Incubation Park**

**03/2024-03/2025**

*AIGC Imaging and Media Entrepreneurial Project* | Co-Founder

- Designed customized image generation workflows based on Stable Diffusion and Midjourney, providing AIGC solutions for industrial clients such as shipping logistics and sports apparel.
- Developed special-effect process for e-commerce model clothing replacement, product rendering, and cosplay photo effects, estimated to cut costs by ~20% and boost efficiency by ~30%.

### **Voluntary Teaching Programme at Maoming Rural Primary School**

**07/2023**

- Designed and delivered interactive lessons for primary school students, while engaging with local communities to gather insights into rural revitalisation and industrial development.
- Wrote a community report and four promotional articles to publicise local industries and rural life, inspiring more students to participate in volunteer education initiatives.

## **AWARDS & HONOURS**

- |                                                                                               |             |
|-----------------------------------------------------------------------------------------------|-------------|
| ➤ Best Poster Award (International), The 25th CICTP Conference, COTA                          | <b>2025</b> |
| ➤ Meritorious Winner (International), Interdisciplinary Contest in Modeling (ICM), COMAP      | <b>2025</b> |
| ➤ Third Prize (Provincial) 18th "Challenge Cup" National Students' Academic Works Competition | <b>2025</b> |
| ➤ Yang Hanlin Scholarship (Top 0.2%, 3/1210), Guangzhou Maritime University                   | <b>2025</b> |
| ➤ Third Prize (Provincial), China Undergraduate Mathematical Contest in Modeling              | <b>2024</b> |
| ➤ Science and Technology Innovation Model Scholarship, Guangzhou Maritime University          | <b>2024</b> |
| ➤ Bronze Award (Provincial), FLTRP · ETIC Cup English Writing Contest                         | <b>2023</b> |

## **SKILLS & LANGUAGES**

**Technical Tools:** Python, MATLAB, SUMO, LaTeX, SPSS, Photoshop, Illustrator

**Languages:** English (fluent, IELTS 6.5), Chinese (native, Mandarin & Cantonese)